

Design and implementation of a 6-DoF robot arm control with object detection based on machine learning using mini microcontroller

Authors: Hayder Hashim Almaliki, Amir Hooshang Mazinan, Seyed Mahmoud Modaresi (2026)

ABSTRACT

This work describes the design and implementation of a 6-DoF robot arm with real-time object detection and pick-and-place capabilities. The control logic is embedded in a resource-constrained AMB82-Mini microcontroller running a quantized YOLOv7-tiny model for object recognition. The system solves analytical inverse kinematics in real time to coordinate joint angles and perform autonomous operations.

RELEVANCE TO THE EB15 ROBOTIC ARM PROJECT

Demonstrates the feasibility of implementing real-time machine learning and advanced kinematics on resource-constrained microcontrollers, validating the use of the ESP32 S3 for high-level supervisor control.

TECHNICAL ANALYSIS & SYSTEM INTEGRATION

This academic research reference documents crucial design paradigms directly relevant to the development of the EB15 six-degrees-of-freedom robotic manipulator. Under the distributed control framework designed for this thesis (featuring the ESP32 S3 as master and the Arduino Uno as slave), the mathematical formulations, communication latencies, and performance profiles analyzed by Hayder Hashim Almaliki provide a benchmark for physical and algorithmic modeling. Specifically, this reference establishes solid validation criteria for timing constraints, closed-loop feedback via absolute magnetic encoders (AS5600), and vibration damping.